Prompt Engineering in Vision-Language Foundation Models: A Detailed Exploration

Jagdish Singh Rathore

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Contents

1 Introduction

Prompt engineering is a technique used to adapt large-scale pre-trained models for downstream tasks without modifying their parameters. In vision-language models (VLMs), prompts are crafted in natural language to describe visual inputs, enabling zero-shot or few-shot learning. This report offers a detailed examination of prompt engineering strategies in general and their implementation in the CLIP model, based on the survey by Gu et al. [2] and the CLIP paper by Radford et al. [5].

2 Prompt Engineering in Large Language Models (LLMs)

Prompt engineering in LLMs is the practice of designing input queries which are known as prompt. In natural language that elicit desired behavior from a pre-trained model like GPT, PaLM without modifying the underlying model weights. This technique became dominant as LLMs demonstrated impressive generalization abilities by using in-context learning. In-context learning means solving unseen tasks using just examples in the form of prompts.

2.1 Common Techniques

- **Zero-shot prompting:** The model is given only the task description, assumes that the model has learned how to perform specific tasks during pre-training, and performance can vary based on phrasing.A direct instruction (e.g., "Translate to German").

- **Few-shot / In-context learning:** The prompt includes a few input-output examples, which are known as shots, that demonstrate the task. No parameters are updated here,Model uses pattern recognition. Stronger performance than Zero-Shot in majority of the cases.

- **Chain-of-Thought (CoT):**Prompt guides the model through intermediate reasoning steps. This encourages more interpretable and accurate output, specifically effective in arithmetic, logic, and multi-hop QA. Decomposes reasoning steps into intermediate steps.

- **ReAct (Reason + Act):** Prompt includes both reasoning and explicit tool use, which is useful in tasks requiring up-to-date information. Combines reasoning with action (e.g., API calls).

2.2 Applications

- Text classification, summarization, translation.

- Open-domain QA and dialogue.

- Code generation and reasoning.

2.3 Strengths and Limitations

- **Advantages:** Prompt Engineering enables LLMs to get adapted for a wide variety of tasks which include translation, summarization, classification, question answering, coding, without the need of fine-tuning. This saves both time and computational resources, specifically when we have to adapt to new domains.

- **Challenges:** Prompt sensitivity, instability across tasks, manual prompt design can be inefficient in some of the cases.

3 Prompt Engineering in Vision-Language Models (VLMs)

Vision language models (VLMs) such as CLIP, FLAMINGO, BLIP, and GPT-4V are capable of processing and understanding both image and text data simultaneously. Prompt engineering plays a vital role in filling the gap between visual content and language, which enables them to perform various downstream tasks in a more meaningful and task-specific manner.

3.1 Techniques

- **Textual Prompting:** In text prompting, the model is guided by using fixed templates of natural language. These prompts align with the input image features with the learned representation in language model enabling classifications without additional training. Templates such as "a photo of a `[class]`" guide classification.

- **Learnable Prompt Embeddings:** Instead of using handcrafted text model like CoOP and CoCoOP, uses trainable embeddings. These embeddings are optimized directly with gradient descent to align the objective of the task. This also allows the prompt vector to adapt for new domains and new datasets, improving performance over manual templates. Methods like CoOp and CoCoOp optimize token embeddings.

- **Multimodal Chain-of-Thought:** Multimodal Chain of Thought technique introduces reasoning through a sequence of steps, trying to mimic how humans analyze visual and textual information together. For example, instead of giving a single answer, the model reasons step-by-step based on both image data and language data, helping with tasks like visual questioning, answering, or complex visual reasoning.Encourages stepwise reasoning over visual and textual inputs.

- **Contextual Visual Prompting:**This method fine-tunes only specific components of a visual language model. For example, the visual encoder, the token embeddings. This helps the model to adapt more quickly and more flexibly to different visual contexts, while keeping its generalization capabilities.

3.2 Applications

- Image classification, VQA (Visual Question Answering), image captioning.

- Visual reasoning, retrieval, and grounded language understanding.

3.3 Strengths and Limitations

- **Advantages:**Effective Zero-Shot Generalization, Vision Language Models such as CLIP and BLIP, can perform various tasks without being explicitly trained on those specific tasks or categories and this is made possible through Effective Prompt Designs. We can reuse vision language models across multiple downstream tasks, such as classification, captioning, and retrieval. This is a cost-effective and scalable approach. The other advantage is faster deployment with less data. Because vision language models rely heavily on pre-training and do not require any fine-tuning for specific tasks, they can be quickly deployed with fewer training examples, ideal for low-resource and domain-adaptive settings.

- **Challenges:** Requires careful cross-modal alignment, lacks maturity compared to LLMs,Sometimes shows biases in Pretraining Data

4 Prompt Engineering in Vision-Language Models

4.1 What is Prompt Engineering?

PROMPT ENGINEERING refers to prompts which are design of input, which elicit specific behaviors from pre-trained model. It was first introduced in Natural Language Processing (NLP) with the rise of large language model(LLMs). Some of the famous models are GPT-3 and DeepSeek[1], where carefully customized input prompts may lead to accurate task performance without training the model on that specific task. In Vision Language Foundation models(VLFMs), this concept was introduced by using natural language descriptions as queries or labels for visual tasks.

prompt engineering in Vision Language Models enables the transformation of the traditional supervised task methods such as image classification into text matching problems. This makes use of language to access the previous model's knowledge in a way that aligns with its specific objective.

4.2 Types of Prompts

- **Hard Prompts:** These are manually written prompts, like "a photo of a giraffe", which describes the visual concept in the most simple language. Advantages of hard prompts are they are easy to implement, but may not generalize as well across tasks or domains. [2].

- **Soft Prompts:** Soft prompts are also known as "prompt tokens". They are nothing but learned continuous vectors added to an input space and trained with a fixed backbone. Soft prompts provide higher flexibility and often perform well as compared to hard prompts, specifically in domain-specific tasks. [3].

- **Hybrid Prompts:** Hybrid prompts are combination of hard and soft components, for example, keeping part of prompt fixed (e.g., “a photo of”) and we optimize the rest which allows leveraging interpretability while side-by-side improving the performance. [2].

4.3 Prompt Placement in Model Architectures

The architecture of VLFMs determines how prompts are integrated:

- **Encoder-only Models (e.g., CLIP):** In this model, prompts are passed through a text encoder and later they are compared to image embeddings, which results that model learns to align the textual description with the visual content.[5].
- **Encoder-Decoder Models (e.g., Flamingo, OFA):** In this model, prompts influence both visual and textual understanding, often conditioning the decoding process.
- **Decoder-only Models (e.g., BLIP-2):** Prompts are often used to generate text autoregressively, which makes them central to their model’s output generation.

4.4 Strategies for Designing Prompts

Several strategies exist for designing effective prompts:

1. **Template Engineering:** Manually writing prompts by using natural language phrases that specify the target concept well. The goal is to select a particular prompt that aligns well with the pre-training data distribution and help the model to get an idea of the task contextually.
2. **Prompt Ensembling:** We use multiple prompt templates for the same class and average their output embeddings to increase and improve robustness because different phrasing may activate different aspects of model-learned representation. In CLIP, prompt ensembling was shown to improve top-1 accuracy significantly on ImageNet. [5].
3. **Prompt Tuning:** Learning prompt embeddings by backpropagation while keeping the rest of the model frozen. For example, instead of text like “a photo of”, soft prompt tuning uses a sequence of learnable vectors to the token sequence.[3].
4. **Prompt Mining:** Prompt mining automatically discovers discrete textual prompts using techniques like gradient-based and research-based techniques. Prompt mining uses token-level optimizations or heuristics to identify which sequence performs better.

4.5 Applications of Prompt Engineering

Prompt engineering supports a wide variety of tasks:

- **Image Classification:** Image classification replaces traditional classifier by using prompt-based matching. (e.g., matching “a photo of a dog” to the image embedding).

- **Object Detection and Segmentation:** Recent methods prompt model for describing a specific region or classifying bounding boxes[2].
- **Visual Question Answering (VQA):** Prompts help form instructive and interpretable queries, which can be used for open-ended tasks.
- **Image Generation and Captioning:** Prompt tuning or conditional prompts steer generative models like DALL-E, Imagen, or BLIP.

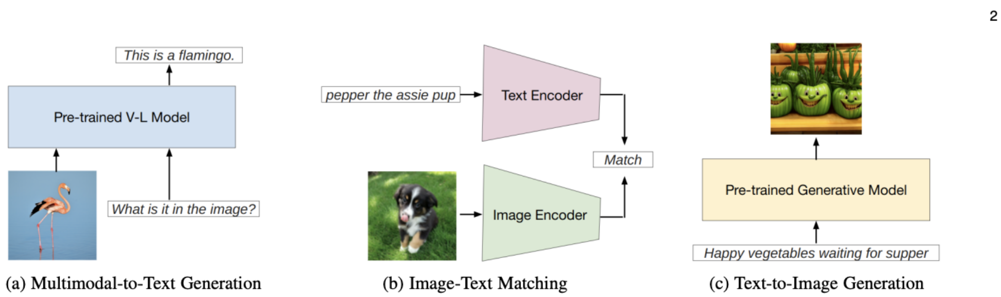

Figure 1: Examples of vision-language model tasks: (a) Multimodal-to-text generation, (b) Image-text matching, and (c) Text-to-image generation [2].

4.6 Challenges and Limitations

Despite its advantages, prompt engineering has notable challenges:

- **Prompt Sensitivity:** Slight variations in wording can lead to significant performance fluctuations.
- **Transferability:** Prompts tuned for one task or dataset may not generalize well.
- **Search Complexity:** The optimal prompt space is vast and difficult to explore exhaustively.
- **Evaluation:** There is limited consensus on how to measure prompt effectiveness fairly across tasks[2].

5 Prompt Engineering in CLIP

5.1 Overview of CLIP

CLIP (Contrastive Language-Image Pretraining) Learns a shared embedding space for text and images. CLIP is trained using a contrastive loss to align image-text pair across a diverse Internet-scalable dataset.[5].

5.2 Role of Prompts in CLIP

Instead of using training classifier, CLIP uses prompts to describe class names and also compute their similarity between text and image embeddings. This makes conversion between classification into a similarity matching task. For example:

- Text: "a photo of a cat"
- Image: Embedding of a cat picture
- Prediction: Highest similarity determines the class

5.3 Prompt Ensembling

As described in above section of the clip paper, Prompt Ensembling involves multiple variations of Prompt. Prompt Ensemblings average their embeddings. This helps to mitigate prompt sensitivity:

- "a photo of a *label*"
- "a blurry photo of a *label*"
- "a cropped photo of a *label*"
- "a black and white photo of a *label*"

How were these prompt templates selected?

The prompt templates used in the research paper were not chosen randomly. They were carefully selected based on specific requirements like visual characteristics and the intended evaluation objective of each dataset.

- **For CIFAR-10:**Templates were designed for the simulation of common image disorders such as low contrast, poor image quality, and the variation in the object size. CIFAR-10 images are often noisy and are small, so this prompt helps the model learn to generalize better across the visual conditions which are degraded[5].
- **For KITTI:** The template directly gives us description about spatial relationship between the observer and the vehicle. Since KITTI majorly focuses on detecting vehicle at various distance, the prompts were aligned with the dataset's core task of distance classification[5].

  KITTI Dataset (Vehicle Distance Classification):

  - "a photo i took of a car on my left or right side."
  - "a photo i took with a car nearby."
  - "a photo i took with a car in the distance."

- **For MNIST:** The dataset majorly consists of grayscale images specifically for handwritten digits. Thus, this simple prompt focused majorly on identifying the digit for effective classification[5].

  MNIST Dataset (Handwritten Digits Recognition):

  – "a photo of the number: {digit}."

Selection strategy across datasets

The strategy involved behind the prompt selection in CLIP involves three important aspects.

1. **Visual Robustness:** The main area of focus is visual robustness, and these templates are chosen to simulate real-world image variation, e.g., contrast shifts, quality differences in the image, blurry images, to improve the robustness against visual distortions which are faced in real-world scenarios.
2. **Task Alignment:** Prompts are carefully aligned and crafted to match the specific prediction goal of each dataset, which ensures that textual descriptions align closely with the inherent tasks.
3. **Descriptive Diversity:** Descriptive diversity means we select multiple prompts to activate different components of the model's learned visual-textual knowledge, which enhances the generalization[5].

Why is this important?

These different designs of prompt templates ensure that CLIP's text encoder effectively captures the full spectrum of visual scenarios present in all the datasets. By using these prompts, CLIP gains several advantages:

1. CLIP becomes more robust to noise and varied image conditions.
2. CLIP improves its ability to generalize to unseen data.
3. CLIP becomes less sensitive to the specific wording of prompts.

Ultimately, prompt ensembling through such carefully selected templates plays a significant role in CLIP's strong zero-shot performance across various vision-language tasks.

5.4 Effectiveness of Prompt Ensembling

CLIP with prompt ensembling achieves significantly better zero-shot performance. On ImageNet, ensembling improves top-1 accuracy from 63.1% (single prompt) to 76.2%[5].

5.5 Limitations and Alternatives

While prompt engineering improves CLIP's performance, it has some limitations:

- **Manual labor:** Requires human creativity and domain knowledge and domain experience
- **Non-optimal:** Manual prompts may not fully capture class semantics.

Recent alternatives like CoOp [6] and ProDA automatically learns prompt vectors through training, providing more robust and transferable results.

5.6 Bias Analysis in CLIP

The main objective was to analyze whether CLIP demonstrates biased behavior towards some specific demographic groups, which was a critical aspect for deploying vision language models correctly in real-world scenarios. They specifically conducted biased analysis by applying CLIP on a dataset which majorly focused on human demographics. One of the key datasets which was used in that is **FairFace**[5], which consists of facial image labeled by attributes such as gender and age.

6 Key Experimental Findings

The CLIP authors use Zero-Shot Classification Experiment to closely examine whether the model's predictions disproportionately associate to a specific demographic group with specific classes.

Their major findings indicate that the clip doesn't exhibit biased patterns, specifically if we focus on the model tends to assign higher prediction scores to some of the demographic groups over others, which demonstrates the demographic imbalance. This behavior is likely inherited from large-scale internet data, which is used during pre-training, which itself contains social and cultural biases.[5]

6.0.1 Prompt Ensembling in Bias Testing

The CLIP documentent does not explicitly mention the use of prompt ensembling during their bias evaluations in Section 7.1. It appears likely that they used a standard prompt such as "a photo of a {label}" for simplicity in their analysis.

The most common reason they chose this option is that the primary goal was to examine the demographic bias in the model's prediction, independent to any other introduced variability due to changing in the prompt phases. By using a consistent prompt, they were able to investigate and check the fairness impacts on a higher degree without having to add more complexity on prompt variation.

However, the authors acknowledge that the model's behavior can vary for different prompts. This suggests that prompt assembly or carefully designing the prompts could potentially influence the fairness outcomes of the model.

7 Proposed Ideas: How Prompt Engineering Can Improve Face Recognition

Based on this finding, there are several strategies that prompt engineering can efficiently improve face recognition systems.

1. **Diverse Prompt Ensembling:** Using a variety of prompts that will describe facial attributes from a different perspectives, such as "a smiling photo of a {label}" or "a close-up portrait of a {label}". This diversity can help to reduce model dependence on the stereotypical descriptions and ultimately improve fairness.

2. **Demographic-Specific Prompt Sets:**Incorporate ROMs that will explicitly mention demographic attributes for promoting balanced model behavior, for example:

   - "a photo of an elderly person with {label}"
   - "a photo of a young adult with {label}"

3. **Fairness-Aware Prompt Optimization:** Develop prompt optimization techniques that will jointly help you to optimize both accuracy and fairness metrics, which will ensure that prompt selection also accounts for demographic, partly, and fairness considerations.

4. **Better Descriptions for Zero-shot Face Recognition: Why it works:** Vision-language models (like CLIP) can match images and text in a shared feature space. Better prompts will eventually lead to better matching.

   Example of Poor Prompt:

   ```
   ''A face of a man.''
   ```

   This is too generic.

   Example of Better Prompt:

   ```
   ''A high-resolution photo of a middle-aged man with short hair,
   wearing black glasses, with a heavy beard, smiling, against a nature
   background.''
   ```

   These approaches will eventually improve fairness, mitigate biases, and also promote responsible development of vision language models in a sensitive application, for example, face recognition.

8 Comparative Insights into Prominent Large Language Models (LLMs)

Model	**Year**	**Params**	**Training Data**	**Special Features / Notes**
GPT-4	2023	~1T	Web, books, code	with Strong reasoning and multimodal (GPT-4o variant)
GPT-4o	2024	Not disclosed	Audio + vision + text	Real-time multimodal can speech input/output
Claude 3	2024	Undisclosed	Constitutional AI	Very strong at math, reasoning, safe
Claude 3.5 Sonnet	2025	Not disclosed	Updated curated datasets	Top-performing model mid-2025
LLaMA 3	2024	8B, 70B	Refined web corpus	Meta's latest open-weight LLM
Mistral 7B	2023	7B	Open web + code	provide high-performance decoder-only
Mixtral 8x7B	2023	12.9B active	Sparse Mixture of Experts	Efficient training with top benchmark results
Gemini 1.5 Pro	2024	Not disclosed	Multimodal + Long-context	Handles over 1M tokens is great for long documents
Gemini 1.5 Flash	2024	Not disclosed	Optimized for speed	Lighter, faster, efficient variant of Pro
Command R+	2024	52B	Retrieval-augmented data	Open-weight RAG-focused model (Cohere)
Phi-3 Mini	2024	3.8B	Small clean datasets	one of the most Compact Microsoft LLM
GLaM	2021	1.2T (MoE)	Curated corpora	A Efficient sparse expert model

Table 1: Comparative table of recent Large Language Models (LLMs).[4]

9 Comparative Insights into Vision-Language Models (VLMs)

Model	Year	Params	Modalities	Special Features / Notes
GPT-4o	2024	Not disclosed	Vision, Audio, Text	Full multimodal input/output incl. live voice
Gemini 1.5 Pro	2024	Not disclosed	Vision and Text	Strong visual understanding with 1M token context
Gemini 1.5 Flash	2024	Not disclosed	Vision and Text	Optimized for latency and also for multimodal inference
Claude 3 Opus	2024	Not disclosed	Vision and Text	Gives High performance vision reasoning
LLaVA-1.5	2023	7B (LLaMA base)	Vision and Text	Open-source and also instruction-tuned visual model
MiniGPT-4	2023	7B	Vision and Text	Has Visual instruction tuning and affordable
InstructBLIP	2023	6.7B	Vision and Text	Strong on instruction-following tasks
Kosmos-2	2023	$\sim$1B	Vision and Text	Grounding + perception also supports OCR
OpenFlamingo	2023	9B–80B	Vision and Text	Open-source Flamingo-style architecture
BLIP-2	2023	$\sim$1B	Vision and Text	Has Two-stage with visual-linguistic fusion
GIT (Microsoft)	2022	345M–2.5B	Vision and Text	Unified vision encoder + language decoder
Flamingo (DeepMind)	2022	80B	Vision and Text	Few-shot image-text inference

Table 2: Comparative table of recent Vision-Language Models (VLMs).[4]

References

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